

Lemma F for the octahedron: opposite petals never overlap

working notes

September 2026 — draft

Audit correction, 9 September 2026. The unrestricted double-outward half-plane claim in the previous draft is false; an integer-coordinate counterexample is checked by rational intervals in `n6/audit_witness.py`. Lemma 5 below is now restricted to the angle range used in Case A. The new base-cone lemma below supplies an exhaustive partition, including the larger angles and equality boundaries: only $\Sigma_W < \pi$ remains after Case A and the base-cone exclusion. The hinge reduction also needs the local nonoverlap hypothesis. Neither these notes nor the numerical experiments prove the full six-vertex theorem. See `n6/REVIEW.md` for the full audit.

Setting and conventions

We use the conventions of `HANDOFF.md`. P is a convex polytope of octahedron type with antipodal (non-adjacent) vertices v, w and equator $u_0, \dots, u_3$ (the neighbours of w , indices mod 4, in counter-clockwise order around w seen from outside). The faces are $W_i = wu_iu_{i+1}$ and $V_i = vu_iu_{i+1}$; κ_x is the curvature at x ; ω_i, ν_i are the angles of W_i at w and of V_i at v . The net Z_k cuts $\text{star}(v) \cup \{wu_k\}$: the fan $W_k, W_{k+1}, W_{k+2}, W'_{k-1}$ around w (angular gap κ_w at the slit) with the petal V_i glued to W_i along u_iu_{i+1} . Touching counts as non-overlapping.

Only nine face pairs of Z_k share no net vertex: the three local pairs at the slit, the two opposite-petal pairs $(V_k, V_{k+2}), (V_{k+1}, V'_{k-1})$, and four petal-vs-far-fan pairs. Lemma F is the statement that under the hypothesis

$$\textbf{(H)} \quad \kappa_v \geq \kappa_x \quad \text{for every vertex } x \text{ of } P \quad (1)$$

none of the six far pairs overlaps, in every Z_k . (The handoff's hypothesis $\kappa_v \geq \max_i \kappa_{u_i}$ is weaker; the numerics below show that the weaker hypothesis is *not* enough for the argument of §3, although no counterexample to the weaker statement itself is known.)

Elementary consequences of (H). Since $\sum_x \kappa_x = 4\pi$ over six vertices, (H) gives $\kappa_v \geq 2\pi/3$, hence $\sum_i \nu_i = 2\pi - \kappa_v \leq 4\pi/3$. At any convex vertex each face angle is at most the sum of the other face angles (the spherical polygon inequality for the tangent cone), so $\nu_j \leq \frac{1}{2} \sum_i \nu_i$, i.e.

$$\nu_j \leq \pi - \frac{1}{2}\kappa_v \leq \frac{2\pi}{3} \quad \text{for every } j. \quad (2)$$

1 Reduction to three consecutive petals

Lemma 1 (petals do not cross interior hinges). *Assume the three local pairs at the slit do not overlap. In Z_k no petal properly crosses the relative interior of a hinge segment wu_j , $j \neq k$.*

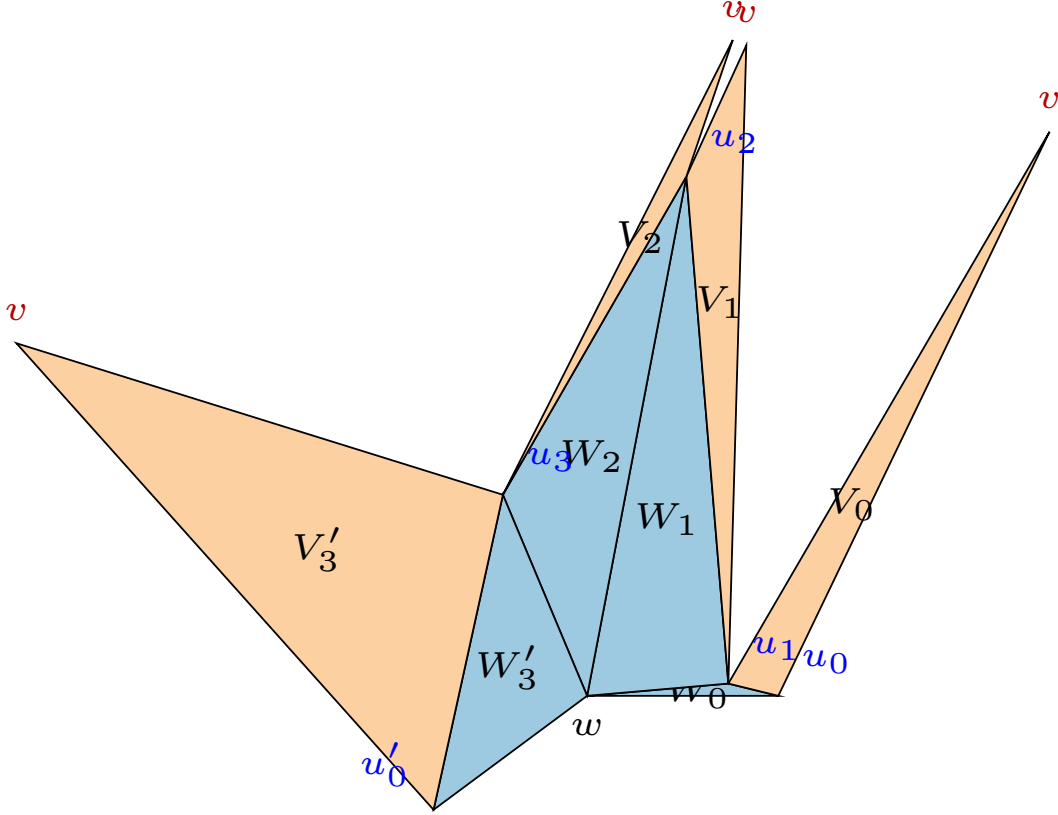


Figure 1: A typical net Z_k (fan faces blue, petals orange), v the sharpest vertex, slit at the sharpest neighbour u_0 of w .

Proof. The two fan faces adjacent to wu_j are W_{j-1} and W_j ; a petal crossing the segment would enter both. Away from the slit, a petal shares a net vertex with W_{j-1} or W_j , and the corresponding wedges are disjoint. There are two exceptions to this shared-copy argument: V'_{k-1} at wu_{k+1} and V_k at wu_{k-1} . In these cases a crossing would enter a local fan face, excluded by the hypothesis. $\square$

Lemma 2 (petal versus far fan face). *Assume the local pairs do not overlap. If a petal V_i overlaps a fan face W_j with $j \notin \{i-1, i, i+1\}$ then V_i overlaps V_j .*

Proof. V_i contains no vertex of W_j (w and the equator vertices of W_j lie in fan faces or petals that share a vertex with V_i , except possibly the local face W'_{k-1} , whose treatment is part of Lemma L). By Lemma 1 the boundary of V_i cannot enter W_j through its hinges, so it enters through the outer edge $u_j u_{j+1}$, which is the common edge of W_j and V_j ; near that crossing V_i meets V_j . $\square$

Subject to Lemma L, Lemma F follows from the following statement about three consecutive petals, applied to the triples (V_k, V_{k+1}, V_{k+2}) and $(V_{k+1}, V_{k+2}, V'_{k-1})$ of Z_k .

Conjecture 3 (Triple Lemma). *Let W_i, W_j, W_{j+1} ($j = i+1$) be three consecutive fan faces of a net and V_i, V_j, V_{j+1} their petals. Under (H), V_i and V_{j+1} do not overlap.*

Numerical evidence 4. No overlap of opposite petals under (H) in 72,000 random nets, nor under the weaker hypothesis $\kappa_v \geq \max \kappa_u$; adversarial hill-climbing (edges ≥ 0.05 diam, curvatures ≥ 0.05) brings the two petals to within 0.4% of the diameter but never into overlap. All observed overlaps (44 in 72,000 nets, without hypothesis) have two equator vertices sharper than v .

2 Where a meeting can be

Fix a triple as in Conjecture 3 and write $u = u_j$, $u' = u_{j+1}$ for the base of V_j , ϕ, ψ for the base angles of V_j at u, u' , $\kappa = \kappa_u + \kappa_{u'}$. In the net, V_i and V_j meet at u only, separated there by the angular gap κ_u between the two copies uv_j and uv_i of the cut edge uv ; likewise at u' . Let L be the line through uv_i and L' the line through $u'v_{j+1}$.

Lemma 5 (double-outward wedge, restricted angle range). *Assume $0 < \phi + \kappa_u < \pi$ and $0 < \psi + \kappa_{u'} < \pi$. V_i lies in the closed half-plane of L not containing V_j , and V_{j+1} in the closed half-plane of L' not containing V_j . Hence every common point of V_i and V_{j+1} lies in $\mathcal{D} :=$ the intersection of these two half-planes, which is the wedge at the point $p^* = L \cap L'$ opposite to the base uu' . If $\kappa < \nu_j$ then $\mathcal{D}$ lies on v_j 's side of the base line (above), and if $\kappa > \nu_j$ it lies on w 's side (below); $\mathcal{D} = \emptyset$ if $\kappa = \nu_j$.*

Proof. The rays uv_i and $u'v_{j+1}$ make angles $\phi + \kappa_u$ and $\psi + \kappa_{u'}$ with the base, so they converge above the base iff $\phi + \psi + \kappa < \pi$, i.e. $\kappa < \nu_j$; the rest is the description of the four wedges at p^* . $\square$

Remark 6. The range assumptions are automatic in Case A, since their sum is $\pi - \nu_j + \kappa < \pi$. They do not follow from (H) in general. Without them even the assertion that V_j is on the opposite side of L from V_i can fail. The rational counterexample in `n6/results/` refutes this unrestricted assertion, not the Case-A proposition.

Lemma 7 (the two cut edges never cross). *In Case A ($\kappa < \nu_j$), the segments uv_i and $u'v_{j+1}$ are disjoint.*

Proof. If they crossed at p then p would be above the base with base angles larger than ϕ, ψ , so v_j lies inside the triangle $uu'p$ and $|up| + |u'p| > |uv_j| + |u'v_j| = |uv_i| + |u'v_{j+1}|$, contradicting $|up| \leq |uv_i|$, $|u'p| \leq |u'v_{j+1}|$. $\square$

Claim 8 (observed overlaps). *In the 44 observed opposite-petal overlaps (no hypothesis), the intersection region lies in $\mathcal{D}$; 26 are above the base and 18 below, exactly as predicted by the sign of $\nu_j - \kappa$. In every below-base case the outer edges uu_i and $u'u_{j+2}$ of W_i, W_{j+1} converge on w 's side (equivalently $\angle_u W_i + \angle_u W_j + \angle_{u'} W_j + \angle_{u'} W_{j+1} < \pi$) and exactly one of the two fan faces extends past the crossing point; both cannot, since fan faces are disjoint.*

3 The rotation picture (above-base case)

Glue V_i, V_j, V_{j+1} flat around v with no gaps (the “ v -fan”); this is possible because $\nu_i + \nu_j + \nu_{j+1} < 2\pi$, and in it the three petals occupy disjoint wedges at v . The net position of V_i is obtained from its v -fan position by the rotation ρ_u about u by the gap angle κ_u (away from V_j), and that of V_{j+1} by the rotation $\rho_{u'}$ about u' by $\kappa_{u'}$. Both rotations have the same sense. Therefore

$$V_i \cap V_{j+1} \neq \emptyset \text{ in the net} \iff \mathcal{R}(V_i^{\text{fan}}) \cap V_{j+1}^{\text{fan}} \neq \emptyset, \quad \mathcal{R} := \rho_{u'}^{-1} \circ \rho_u,$$

and $\mathcal{R}$ is the rotation by $\kappa = \kappa_u + \kappa_{u'}$ about the point c^* on w 's side of the base with $\angle c^*uu' = \kappa_u/2$, $\angle uu'c^* = \kappa_{u'}/2$ (composition of two rotations). Let M be the line c^*v ; $\mathcal{R}$ moves points from u' 's side of M towards u 's side. Measure angles around c^* from M , positively towards u 's side.

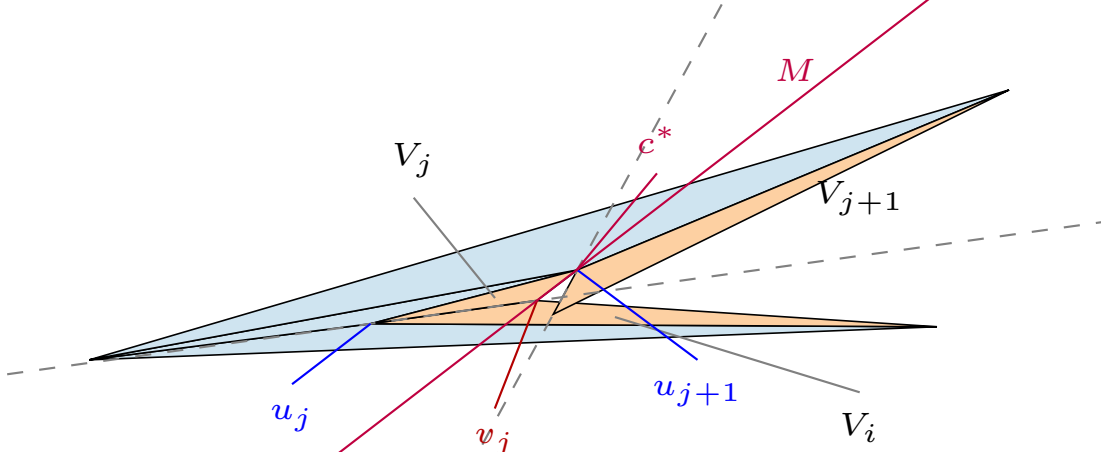


Figure 2: A thin above-base overlap (no hypothesis): the dashed cut-edge lines, the point c^* and the line M of §3. Here $\nu_i + \nu_j > \pi$ and V_i crosses M .

Lemma 9 (angular criterion). *Let $\varepsilon_i \geq 0$ be the largest angle (seen from c^*) by which $V_i^{\text{fan}} \cap \{\text{above the base}\}$ crosses M towards u' 's side, and ε_{j+1} the corresponding excursion of $V_{j+1}^{\text{fan}} \cap \{\text{above}\}$ towards u 's side. If $\kappa < \nu_j$ and*

$$\varepsilon_i + \varepsilon_{j+1} < \kappa, \quad (3)$$

then V_i and V_{j+1} do not overlap in the net.

Proof. By Lemma 5 a common point p is above the base. The rotation ρ_u^{-1} maps the wedge of V_i at u in the net, $[\phi + \kappa_u, 2\pi - \angle_u W_j - \angle_u W_i]$, onto the v -fan wedge, so $x = \rho_u^{-1}(p)$ is below the base only if p is; hence $x \in V_i^{\text{fan}}$ and $y = \rho_{u'}(p) \in V_{j+1}^{\text{fan}}$ are above the base and $y = \mathcal{R}(x)$. Their angles from M satisfy $\theta_y = \theta_x + \kappa$ with $\theta_x \geq -\varepsilon_i$ and $\theta_y \leq \varepsilon_{j+1}$, so $\kappa \leq \varepsilon_i + \varepsilon_{j+1}$ (no wrap-around since $\kappa < \nu_j < \pi$). $\square$

In particular, if neither petal crosses M above the base ($\varepsilon_i = \varepsilon_{j+1} = 0$), there is no overlap. When does a petal cross M ? Write $\mu_u = \angle uv c^*$, $\mu_{u'} = \angle c^* v u'$, so $\mu_u + \mu_{u'} = \nu_j$ (c^* lies in the cone of V_j at v , being on w 's side of the base between the lines vu, vu'). The wedge of V_i at v starts at vu and turns away from V_j by ν_i ; it reaches the ray of M beyond v iff $\nu_i > \pi - \mu_u$. The other way for $V_i^{\text{fan}} \cap \{\text{above}\}$ to cross M is through the far vertex u_i or through the point where the edge vu_i or uu_i crosses the base line.

Lemma 10. *Under (H) the two petals cannot both cross M through their wedges at v .*

Proof. That would need $\nu_i + \mu_u > \pi$ and $\nu_{j+1} + \mu_{u'} > \pi$, hence $\nu_i + \nu_{j+1} + \nu_j > 2\pi$, whereas $\nu_i + \nu_j + \nu_{j+1} \leq \sum \nu \leq 4\pi/3$. $\square$

Lemma 11 (the above-base case needs a very sharp v). *Under (H), $\kappa < \nu_j$ forces $\kappa_v > 6\pi/7$, hence $\sum_i \nu_i < 8\pi/7$ and each $\nu_i < 4\pi/7$.*

Proof. $4\pi = \sum_x \kappa_x \leq 4\kappa_v + \kappa$ and $\kappa < \nu_j \leq \pi - \kappa_v/2$ by (2). $\square$

Numerical evidence 12. In 3000 random octahedra (all apexes, all triples with $\kappa < \nu_j$): the composition $\mathcal{R}$ has the stated centre and sense, and the criterion of Lemma 9 reproduces the net overlap test exactly. Under (H) neither petal ever crosses M above the base ($\varepsilon_i = \varepsilon_{j+1} = 0$ in all 224 triples, degenerate ones included), and adversarial search under (H) cannot make either excursion

positive. Under the weaker hypothesis $\kappa_v \geq \max \kappa_u$ a petal can cross M and (3) can fail (excursion 1.96 with $\kappa = 0.05$), so (H) is genuinely used. Every one of the 26 observed above-base overlaps violates $\varepsilon_i = \varepsilon_{j+1} = 0$ (20 through V_i , 6 through V_{j+1}).

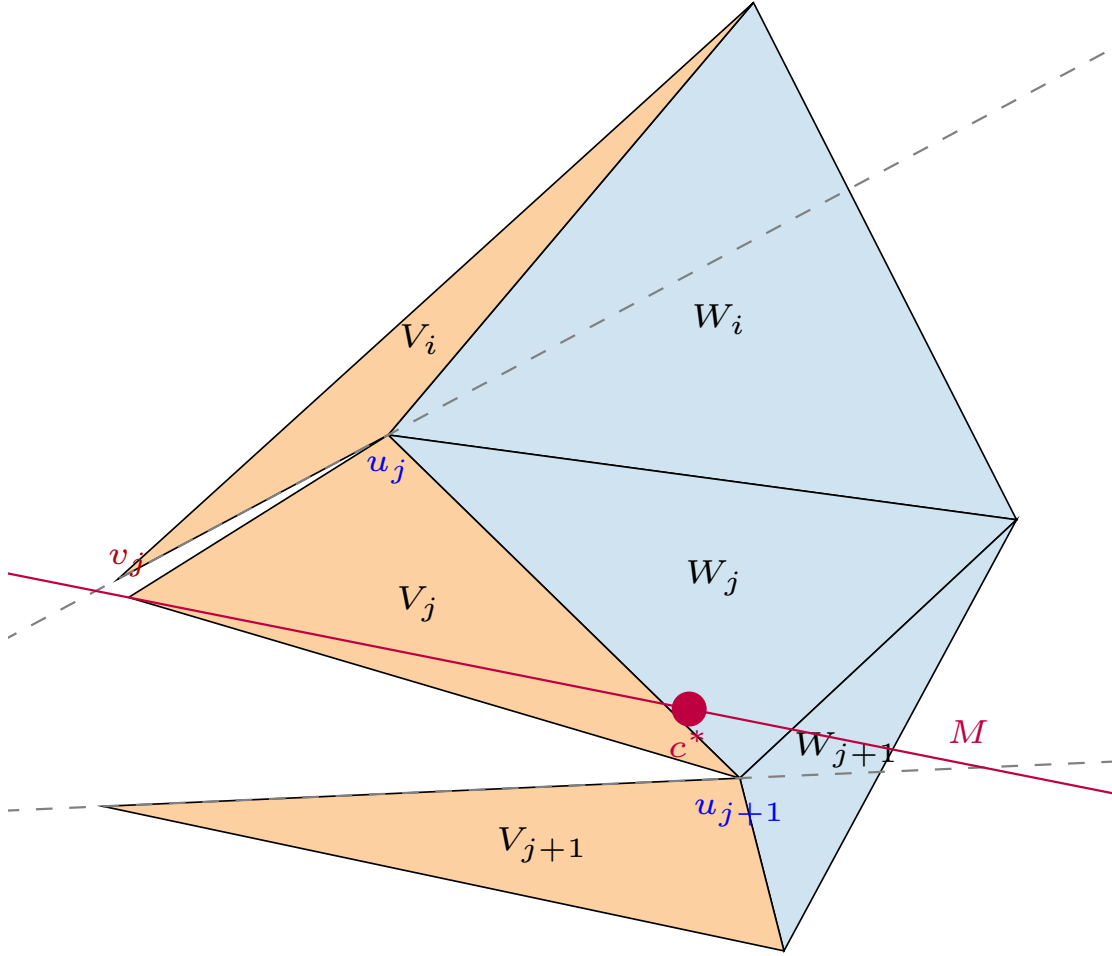


Figure 3: A triple under (H) with $\kappa < \nu_j$: c^* , the line M and the two dashed cut-edge lines; both petals stay on their own side of M .

Superseded: what remained for the rotation argument. Prove that under (H) and $\kappa < \nu_j$ neither $V_i^{\text{fan}} \cap \{\text{above}\}$ nor $V_{j+1}^{\text{fan}} \cap \{\text{above}\}$ crosses M (or, more weakly, that (3) holds). The three crossing mechanisms are: (a) the wedge at v : $\nu_i > \pi - \mu_u$; (b) the far vertex u_i above the base on u' 's side of M ; (c) an edge of V_i crossing the base line outside the segment uu' on u' 's side of M . Mechanism (c) through vu_i requires $\nu_i + \nu_j > \pi + \psi$; (a) requires $\nu_i + \nu_j > \pi$; by Lemma 11 these need $\kappa_v < \pi$ and two of the four angles at v close to $4\pi/7$. The numerics say the fan faces then obstruct this, but the argument is not yet written.

4 Case A is closed: apex cones

The side conditions of §3 are not needed. Two lemmas finish the above-base case.

Lemma 13 (no wrap). *Under (H), $\nu_i + \nu_j \leq \pi + \kappa_u + \kappa_{u'}$ and $\nu_{j+1} + \nu_j \leq \pi + \kappa_u + \kappa_{u'}$ for every triple.*

Proof. From $\sum_x \kappa_x = 4\pi$ and (H), $\kappa := \kappa_u + \kappa_{u'} = 4\pi - \kappa_v - \kappa_w - \kappa_{u_i} - \kappa_{u_m} \geq 4\pi - 4\kappa_v = 4\sum \nu - 4\pi$. If $\sum \nu \geq \pi$ then $\pi + \kappa \geq 4\sum \nu - 3\pi \geq \sum \nu \geq \nu_i + \nu_j$; if $\sum \nu < \pi$ then $\nu_i + \nu_j < \pi < \pi + \kappa$. $\square$

Lemma 14 (apex cones). *Let C_i be the closed cone at v_i spanned by the cut edges $v_i u$ and $v_i u_i$ (it contains V_i), and C_{j+1} the cone at v_{j+1} spanned by $v_{j+1} u'$ and $v_{j+1} u'_m$. If $\kappa < \nu_j$, $\nu_i \leq \pi - \nu_j + \kappa$ and $\nu_{j+1} \leq \pi - \nu_j + \kappa$, then C_i and C_{j+1} have disjoint interiors. In particular V_i and V_{j+1} do not overlap.*

Proof. Measure directions counter-clockwise from $u \rightarrow u'$ and put $\alpha = \phi + \kappa_u$, $\beta = \psi + \kappa_{u'}$, so $\alpha + \beta = \pi - \nu_j + \kappa < \pi$. The cut edge uv_i has direction α (the gap turns v_j away from V_j) and V_i lies on its left, so from v_i the cone C_i is the sector $I_i = [\pi + \alpha - \nu_i, \pi + \alpha]$ ending at the direction e_1 towards u . Likewise $u'v_{j+1}$ has direction $\pi - \beta$, V_{j+1} lies on its right, and C_{j+1} is the sector $I_{j+1} = [-\beta, -\beta + \nu_{j+1}]$ starting at the direction towards u' . Two cones meet iff $d := v_{j+1} - v_i$ lies in the convex cone K generated by $I_i \cup (-I_{j+1})$, where $-I_{j+1} = [\pi - \beta, \pi - \beta + \nu_{j+1}]$. The two hypotheses say exactly that I_i and $-I_{j+1}$ lie inside $S = [\pi - \beta, \pi + \alpha]$, a sector of width $\alpha + \beta < \pi$ bounded by e_1 and by $f_2 =$ the direction of $u' \rightarrow v_{j+1}$; hence $K = S$. Now $d = s_u e_1 + \ell e_0 + s_{u'} f_2$ with e_0 the base direction, and writing $e_0 = c_1 e_1 + c_2 f_2$ gives $c_1 = -\sin \beta / \sin(\alpha + \beta)$, $c_2 = -\sin \alpha / \sin(\alpha + \beta)$. Thus $d = (s_u - |up^*|)e_1 + (s_{u'} - |u'p^*|)f_2$ where p^* is the crossing of the two cut-edge lines, and $d \in S$ iff $s_u \geq |up^*|$ and $s_{u'} \geq |u'p^*|$, i.e. iff both cut edges reach p^* , i.e. iff they cross. Lemma 7 excludes this (equality means the cones touch at p^*). $\square$

Proposition 15. *Under (H) the Triple Lemma holds whenever $\kappa_u + \kappa_{u'} < \nu_j$ (Case A).*

Proof. Lemma 13 gives the two angle conditions of Lemma 14. $\square$

Numerical evidence 16. On 3241 random Case-A triples without hypothesis the cones never meet when the two angle conditions hold, and the conditions fail in 8% of them; under (H) they never fail (208 triples; adversarial margin 1.3 rad, `adv_wrap.py`), and the cone intersection has empty interior in every Case-A triple (adversarial depth -0.0007 diameters at the curvature guard, `adv_cones.py`). Without hypothesis the adversary makes the cones meet (depth $+0.006$), as it must, since 26 above-base overlaps exist.

Case B in this language. Any meeting lies in $C_i \cap C_{j+1}$, and in Case B this intersection can be non-empty. Cones at the two base vertices, rather than at the apexes, give the additional exclusion below.

5 An exhaustive partition: the base cones

Put $u = (0, 0)$, $u' = (\ell, 0)$, with the middle petal above the base and the middle fan face below it. Write

$$\alpha = \phi + \kappa_u, \quad \beta = \psi + \kappa_{u'}, \quad \sigma = \angle_u W_i + \angle_u W_j, \quad \tau = \angle_{u'} W_j + \angle_{u'} W_{j+1},$$

and $a = \angle_u V_i$, $b = \angle_{u'} V_{j+1}$. Vertex angle sums give

$$\alpha + \sigma + a = 2\pi, \quad \beta + \tau + b = 2\pi, \quad \alpha + \beta = \pi - \nu_j + \kappa.$$

All four angles $\alpha, \beta, \sigma, \tau$ are in $(0, 2\pi)$, and $a, b \in (0, \pi)$. The actual base cones of the outer petals have direction intervals $I = [\alpha, 2\pi - \sigma]$ at u and $J = [\pi + \tau, 3\pi - \beta]$ at u' . These are unwrapped intervals of widths $a, b < \pi$, valid even if α or β reaches or exceeds π .

Lemma 17 (base cones). *If $\alpha + \beta \geq \pi$ and $\sigma + \tau \geq \pi$, the two base cones are disjoint. No curvature-ranking hypothesis is required.*

Proof. The left cone and the reversed right cone have their directions in $[\alpha, 2\pi - \sigma]$ and $[\tau, 2\pi - \beta]$, respectively. Both are contained in $S = [m, 2\pi - n]$, where $m = \min(\alpha, \tau) > 0$ and $n = \min(\sigma, \beta) > 0$. Since $a, b < \pi$,

$$m + n = \min(\alpha + \sigma, \alpha + \beta, \tau + \sigma, \tau + \beta) \geq \pi.$$

Thus S is a closed convex sector of width at most π , excluding the positive base direction. A meeting of the two translated cones would express $u' - u = (\ell, 0)$ as a vector in the left cone minus one in the right cone. That difference belongs to S , a contradiction. Equality is included. $\square$

Proposition 18 (complete partition). *Under (H), only $\Sigma_W := \sigma + \tau < \pi$ remains to prove for the opposite-petal statement. This regime automatically has $\kappa > \nu_j$. Neither $\kappa = \nu_j$ nor $\Sigma_W = \pi$ leaves a boundary case open.*

Proof. If $\kappa < \nu_j$, Case A applies. Otherwise $\alpha + \beta \geq \pi$, so Lemma 17 excludes a meeting when $\Sigma_W \geq \pi$. Finally $\alpha + \beta + \Sigma_W = 4\pi - a - b > 2\pi$, so $\Sigma_W < \pi$ implies $\alpha + \beta > \pi$, hence $\kappa > \nu_j$. When $\kappa = \nu_j$, the same identity gives $\Sigma_W > \pi$. $\square$

Remark 19. An exact integer-coordinate example in `n6/CASE_PARTITION.md` has (H), the sharpest-neighbour slit rule, and $\alpha > \pi$. Rational interval checks certify its curvature ordering and angular regime. Its fan-angle sum is above π , so Lemma 17 handles it. The unrestricted D-lemma remains false; it is not needed for the remaining outer-wedge argument.

Lemma 20 (remaining apex test). *In the remaining region $\Sigma_W = \sigma + \tau < \pi$, set $s = |vu|$, $s' = |vu'|$, and $\ell = |uu'|$. The left apex is strictly beyond the right outer fan-edge line, on the side away from w , if and only if*

$$\Sigma_W + a < \pi \quad \text{and} \quad s \sin(\Sigma_W + a) > \ell \sin \tau.$$

The corresponding right-apex test is

$$\Sigma_W + b < \pi \quad \text{and} \quad s' \sin(\Sigma_W + b) > \ell \sin \sigma.$$

Proof. The left apex is $s(\cos \alpha, \sin \alpha)$, while the right outer ray has direction $\pi + \tau$ from $(\ell, 0)$. The signed determinant, positive on the side away from w , is $s \sin(\tau - \alpha) - \ell \sin \tau$. The angle sums give $\alpha - \tau = 2\pi - \Sigma_W - a \in (0, 2\pi)$, so $\sin(\tau - \alpha) = \sin(\Sigma_W + a)$; positivity requires $\Sigma_W + a < \pi$. Reflection gives the right test. $\square$

Remark 21. The sine rule gives $s/\ell = \sin \psi / \sin \nu_j$ and $s'/\ell = \sin \phi / \sin \nu_j$. These are exact reformulations, not proofs of non-entry under the selection rules. The stronger candidate bounds $\Sigma_W + a \geq \pi$, $\Sigma_W + b \geq \pi$ have only numerical support. An exact abstract angle countermodel in `n6/CASE_PARTITION.md` shows that triangle sums, cone inequalities, and curvature rankings alone cannot prove these stronger bounds; metric realizability information is needed.

6 The below-base case (Case B)

By Proposition 18, it remains to consider $\Sigma_W < \pi$. This implies strict Case B and places both individual fan-angle sums below π . No restriction on $\phi + \kappa_u$ or $\psi + \kappa_{u'}$ is imposed here. For consistency with the historical formulas in this section, rename the two fan-angle sums α, β (they were σ, τ above). Write $u_k = u_{j+2}$ for the far vertex of W_{j+1} (in Z_k this is the slit copy, $k = i - 1$), A for the line through the outer edge uu_i of W_i , C for the line through $u'u_k$, $\alpha = \angle_u W_i + \angle_u W_j$, $\beta = \angle_{u'} W_j + \angle_{u'} W_{j+1}$ and $\Sigma_W = \alpha + \beta$ (the four fan angles at u, u'). Let $\kappa_{\text{back}} = \kappa_{u_i} + \kappa_{u_k} + \kappa_w$ be the total gap on the slit side, so that $\kappa_v + \kappa + \kappa_{\text{back}} = 4\pi$, and let $V_m = V_{i-1}$ be the fourth petal, glued to $W_m = W_k$. Throughout the remaining wedge analysis, $0 < \alpha, \beta$ and $\alpha + \beta = \Sigma_W < \pi$.

Lemma 22 (the wedge Ω). *V_i lies in the closed half-plane H_A of A not containing w , and V_{j+1} in the half-plane H_C of C not containing w ; every common point lies in $\Omega := H_A \cap H_C$. The lines cross at a point X on w 's side of the base, w lies in the triangle $uu'X$, and Ω is the wedge at X opposite to that triangle, of angle $\theta_X = \pi - \Sigma_W$.*

Proof. V_i and W_i lie on opposite sides of their common edge line A . On the base line, H_A is the ray beyond u away from u' and H_C the ray beyond u' away from u , so Ω misses the base line and lies on one side of it. The rays $u \rightarrow u_i$ and $u' \rightarrow u_k$ make the angles α, β with the base on w 's side; they converge there since $\Sigma_W < \pi$, and X and Ω are below the base. w is on u' 's side of A , on u 's side of C and below the base. $\square$

Lemma 23 (how a petal reaches Ω). *Let $\Sigma_W < \pi$. V_i meets the interior of Ω iff u_i lies past X on A (far vertex, f) or $v_i \in \text{int } \Omega$ (apex, a); likewise V_{j+1} with u_k and v_{j+1} . The far vertices u_i, u_k are never both past X .*

Proof. $V_i \subset H_A$ meets the open half-plane of C iff one of its vertices does; u is on w 's side of C , and $u_i \in A$ is beyond C iff it is past X . If both far vertices were past X , then $X \in uu_i \cap u'u_k \subset W_i \cap W_{j+1} = \{w\}$, but $w \notin A$. $\square$

So a meeting needs $\Sigma_W < \pi$ and one of the sub-cases (a, a), (fa, a), (a, fa), (f, a), (a, f): at least one apex lies in Ω .

Lemma 24 (direction condition). *For every triple,*

$$\angle_{u_k} V_{j+1} - \Sigma_W = \nu_i + \nu_m + \angle_u V_i - \kappa_{\text{back}}, \quad \angle_{u_i} V_i - \Sigma_W = \nu_{j+1} + \nu_m + \angle_{u'} V_{j+1} - \kappa_{\text{back}}.$$

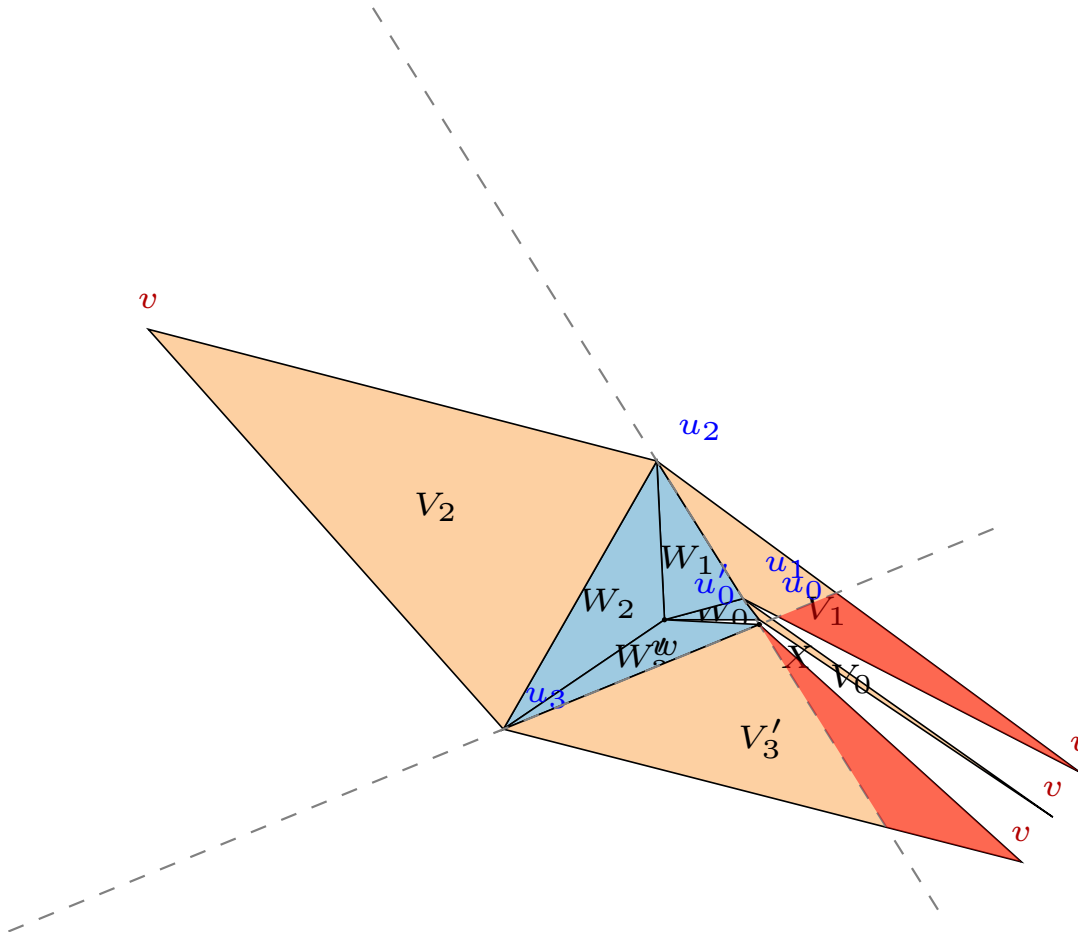
If $\Sigma_W < \pi$, u_k is strictly before X , and v_{j+1} is in the interior of Ω , then $\angle_{u_k} V_{j+1} > \Sigma_W$. The mirror statement requires u_i strictly before X . No such direction conclusion is asserted here when the respective far vertex is already past X .

Proof. $\alpha = 2\pi - \kappa_u - \phi - \angle_u V_i$, $\beta = 2\pi - \kappa_{u'} - \psi - \angle_{u'} V_{j+1}$, $\phi + \psi = \pi - \nu_j$ and $\angle_{u_k} V_{j+1} = \pi - \angle_{u'} V_{j+1} - \nu_{j+1}$ give $\angle_{u_k} V_{j+1} - \Sigma_W = \kappa + \angle_u V_i - \nu_j - \nu_{j+1} - 2\pi$, and $\kappa = 2\pi + \sum \nu - \kappa_{\text{back}}$. For the direction: $u_k \in C$ lies on w 's side of A and $v_{j+1} \in \Omega$, so the direction $u_k \rightarrow v_{j+1}$ points to the far side of C (the open half-circle of directions on V_{j+1} 's side of C) and across A ; the direction of A towards u_i makes the angle Σ_W with $u_k \rightarrow u'$ (exterior angle at X), so the admissible directions are those making an angle in (Σ_W, π) with $u_k \rightarrow u'$, and $u_k \rightarrow v_{j+1}$ makes the angle $\angle_{u_k} V_{j+1}$. $\square$

Lemma 25 (sub-case (a,a): the hexagon identity). *Let $\Sigma_W < \pi$, neither far vertex past X , $v_i, v_{j+1} \in \Omega$, and assume the three local pairs at the slit do not overlap (Lemma L). If p is a common point of V_i and V_{j+1} , then*

$$\nu_m = \kappa_{\text{back}} + \gamma_p + \angle(p u_i v_i) + \angle(p u_k v_{j+1}), \quad \gamma_p > 0,$$

Proof. The closed curve $\Gamma = u_i \rightarrow p \rightarrow u_k \rightarrow w \rightarrow u_i$ (segments in $V_i, V_{j+1}, W_{j+1}, W_i$) is simple: $[w, u_k]$ lies on w 's side of A and meets $V_i \subset H_A$ at most at X , symmetrically for $[w, u_i]$. It lies in the triangle $\Delta = uu'p$, which is on w 's side of the base line because $p \in \Omega$. Let Q be the region bounded by Γ . Its angle at w is $\omega_m + \kappa_w$ or $\omega_i + \omega_j + \omega_{j+1}$; in the second case W_j , hence V_j (which meets Γ nowhere, by the shared-vertex lemma), lie in $Q \subset \Delta$, but v_j is above the base. So Q contains W_m near w , hence W_m and V_m (their boundaries do not cross Γ : shared vertices with V_i, W_i, W_{j+1} , and Lemma L for the pairs with V_{j+1}, W_{j+1}). The hexagon $R = Q \setminus (W_m \cup V_m)$ has interior angles $\angle_{u_i} V_i - \angle(p u_i u) + \kappa_{u_i}$ at u_i , γ_p at p , $2\pi - \angle_{u_k} W_{j+1} - \angle(p u_k u')$ at u_k , κ_w at w , $2\pi - \angle_{u'_k} W_m - \angle_{u'_k} V_m$ at u'_k and $2\pi - \nu_m$ at v_m ; their sum is 4π , and $2\pi - \angle_{u_k} W_{j+1} - \angle_{u'_k} W_m - \angle_{u'_k} V_m = \kappa_{u_k} + \angle_{u_k} V_{j+1}$. $\square$



Regime split for (a,a). If $\nu_m \leq \kappa_{\text{back}}$ Lemma 25 excludes a meeting. If $\nu_m > \kappa_{\text{back}}$, the two outer cut edges $u_i v_i$ and $u_k v_{j+1}$ make the angles $\angle_{u_i} V_m + \kappa_{u_i}$ and $\angle_{u'_k} V_m + \kappa_{u_k} + \kappa_w$ with the segment $u_i u_k$, so their lines converge on V_m 's side with angle $\nu_m - \kappa_{\text{back}}$, and Lemma 14 applies verbatim with the base uu' replaced by $u_i u_k$ (no-wrap $\nu_i, \nu_{j+1} \leq \pi - \nu_m + \kappa_{\text{back}}$ follows from (H))

as in Lemma 13): the cones C_i, C_{j+1} have disjoint interiors unless the two outer cut edges cross. Two gaps remain here: (i) a no-cross lemma for the outer cut edges across the slit ($u_k \neq u'_k$; the argument of Lemma 7 needs $|u_k v_m| \geq |u'_k v_m|$); (ii) the sub-cases with a far vertex past X .

Lemma 26 (the slit rule excludes the large-back-angle alternative). *If u_k has maximum curvature among the four equator vertices, then for every face angle ν_t at v ,*

$$\kappa_{u_k} + \kappa_w - \nu_t \geq \frac{1}{4}\kappa_v + \frac{3}{4}\kappa_w > 0.$$

In particular $\kappa_{\text{back}} > \nu_m$. Thus the (a,a) sub-case is excluded by Lemma 25, conditional on Lemma L. No sharpest-apex assumption is needed for this bound.

Proof. Gauss–Bonnet and the slit rule give $4\pi \leq \kappa_v + 4\kappa_{u_k} + \kappa_w$, while the convex-vertex cone inequality gives $\nu_t \leq \pi - \kappa_v/2$. Subtract these bounds, and use strict positivity of both curvatures. $\square$

Numerical evidence 27. Under (H), in 23,578 Case-B triples, 19,858 have $\Sigma_W \geq \pi$ and only 27 have both petals reaching Ω (sub-cases (a,a) 22, (fa,a) 4, (a,fa) 1); no (f,f), as it must be. Seeded adversarial closest approach inside Ω (`caseB_sub.py adv`): (a,a) 3.5%, (fa,a) 3.2%, (a,fa) 1.9% of the diameter (the last slightly past the guard). Without hypothesis (a,fa) overlaps (+1.6%) and (a,a) stays 0.7% apart. In every (a,a) triple with $\nu_m > \kappa_{\text{back}}$ (109 found, with and without (H)) the apex cones have disjoint interiors, and the outer cut edges never cross when $\nu_m > \kappa_{\text{back}}$ (adversarial best -0.003 diameters, `adv_crossA.py`); $\nu_m > \kappa_{\text{back}}$ does occur under (H) in (a,a) (excess up to 0.30 rad), so both regimes are needed. An earlier count “15% of Case-B triples have both petals reaching the cone intersection” used a vacuous test for the position of X and is withdrawn.

Historical numerical evidence for the slit rule. The old `adv_angle.py` mode i used the wrong base vertex; its saved margins are withdrawn for the intended far-angle claim. The code is corrected, but the original searches have not all been rerun. The direct half-plane probes below are different experiments. Lemma F is used with the slit at the sharpest neighbour u_k of w ($\kappa_{u_k} \geq \kappa_{u_t}$ for all t ; Lemma L needs this rule: without it the local pairs overlap under (H) in 3.4% of nets). Under (H) and the rule, in 641 Case-B triples with $\Sigma_W < \pi$ the petal V_i never reaches Ω and V_{j+1} reaches it only through its far vertex; adversarially (`adv_reach.py`, `adv_angle.py`): v_i stays 0.35 diameters short of C (with $\Sigma_W \leq \pi - 0.3$), v_{j+1} stays 0.12 diameters short of A , u_i stays 0.026 short of X , and under the rule alone $\angle_{u_k} V_{j+1} \leq \Sigma_W - 0.35$. By Lemma 23 it therefore suffices to prove, under (H) and the rule, that neither apex lies in Ω : **(P1)** $v_{j+1} \notin \Omega$, **(P2)** $v_i \notin \Omega$. Linear programming over the 24 face angles (face sums, cone inequalities, curvature bounds, the rule and (H)) numerically allows $\angle_{u_k} V_{j+1} - \Sigma_W$ up to $\pi/2$. This suggests that these linear constraints alone are insufficient; additional metric or angle-realizability constraints may be needed. The tight sampled configurations are needle-shaped (Figure above). This is the recommended route: it replaces the five sub-cases by two apex statements within the stated angle regime. It now applies to the complete remaining region $\Sigma_W < \pi$ by Proposition 18. It does not settle those two apex statements or Lemma L. Lemma 26 separately removes the (a,a) branch under the slit rule, conditional on the local nonoverlap required by the hexagon lemma.

7 Known false / do not retry

Simple separating lines for the two petals inside $\mathcal{D}$ (bisector of $\mathcal{D}$ at p^* , lines through v_j) fail in about a third of the non-degenerate cases even under (H): the separation is a rotation phenomenon,

not a half-plane one. The six wedge-lean inequalities of the handoff fail in 0.5% of nets under (H) (needle-like w), so they cannot replace the argument above.

Code

`octa.py` (module: all quantities, nets Z_k , nine pairs, intersection polygons), `vwedge.py`, `oppcases.py` (collect/classify/draw overlaps), `adv_opp.py` (adversarial separation), `cstar.py` (checks of §3), `adv_eps.py`, `adv_mu.py`, `adv_cstar.py` (adversarial tests of (3) and of the crossing conditions), `tikz_net.py` (figures), `draw.py` (PNG); Case B: `caseB_sub.py` (sub-cases, seeded adversary), `caseB_draw.py`, `adv_crossA.py`, `adv_reach.py`, `adv_angle.py`, `adv_rule.py`, `lp_angles.py`.